GitHub link: https://github.com/Ramya2003sri/meachine-learning-and-neural-networks

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Module: Machine Learning and Neural Networks

Customer Segmentation Using K-Means Clustering: Study of credit card

1. Introduction

Clustering is an unsupervised machine learning technique used to group similar data points without prior labels. It helps uncover hidden patterns in data.

This project applies K-Means clustering to a credit card dataset to identify distinct customer segments based on their financial behaviour. The goal is to support business decision-making such as targeted marketing and risk management.

In this tutorial K-Means clustering is applied to a credit card dataset containing information about customer transactions, balances, and payment behaviour. The aim is to uncover underlying patterns in customer activity and group individuals into meaningful segments. These segments can then be used to support data-driven decision-making, such as identifying high-value customers, detecting low-engagement users, and improving risk assessment strategies.

2.What is K-Means Clustering?

K-Means is an unsupervised machine learning algorithm used to group data into a predefined number of clusters based on similarity. The goal of the algorithm is to partition data such that points within the same cluster are more similar to each other than to those in other clusters. The algorithm works by minimising the within-cluster variance, often referred to as inertia, which is the sum of squared distances between data points and their assigned cluster centroids. Similarity is typically measured using Euclidean distance, making K-Means sensitive to the scale of input features. K-Means operates through an iterative process: 1. Initialisation: Select K initial centroids (either randomly or using methods such as kmeans) 2. Assignment step: Assign each data point to the nearest centroid based on distance 3. Update step: Recalculate each centroid as the mean of all points assigned to that cluster 4. Repeat: Continue the assignment and update steps until the centroids no longer change significantly or a maximum number of iterations is reached. Despite its simplicity, K-Means is powerful and efficient for large datasets. However, it requires the number of clusters (K) to be specified in advance and is sensitive to feature scaling and initial centroid placement, which can affect the final clustering outcome.

3.Focus of This Tutorial Although

K-Means is a simple and widely used clustering algorithm, its effectiveness depends on several important factors. This tutorial focuses on three key aspects that influence its performance. First, the role of feature scaling is examined, as K-Means relies on distance-based calculations and is sensitive to differences in feature magnitude. Second, the process of selecting an appropriate number of clusters is explored using the Elbow Method, which helps identify a suitable value of K based on clustering performance. Finally, Principal Component Analysis (PCA) is applied to reduce the dimensionality of the dataset and enable visualisation of the clustering results. By focusing on these aspects, this tutorial aims to demonstrate how K-Means can be applied more effectively and how its results can be better interpreted in practical scenarios.

4. Objectives

- To analyse customer credit card usage patterns
- To group customers into meaningful clusters
- To evaluate clustering performance using appropriate metrics
- To visualise and interpret customer segments

5. Dataset Description

The dataset contains credit card usage information for customers, including:

- **Balance** – Remaining account balance
- **Purchases** – Total purchase amount
- **Credit Limit** – Maximum credit allowed
- **Payments** – Amount paid by the customer

6. Methodology

6.1 Data Preprocessing

Since K-Means is distance-based, feature scaling is essential.

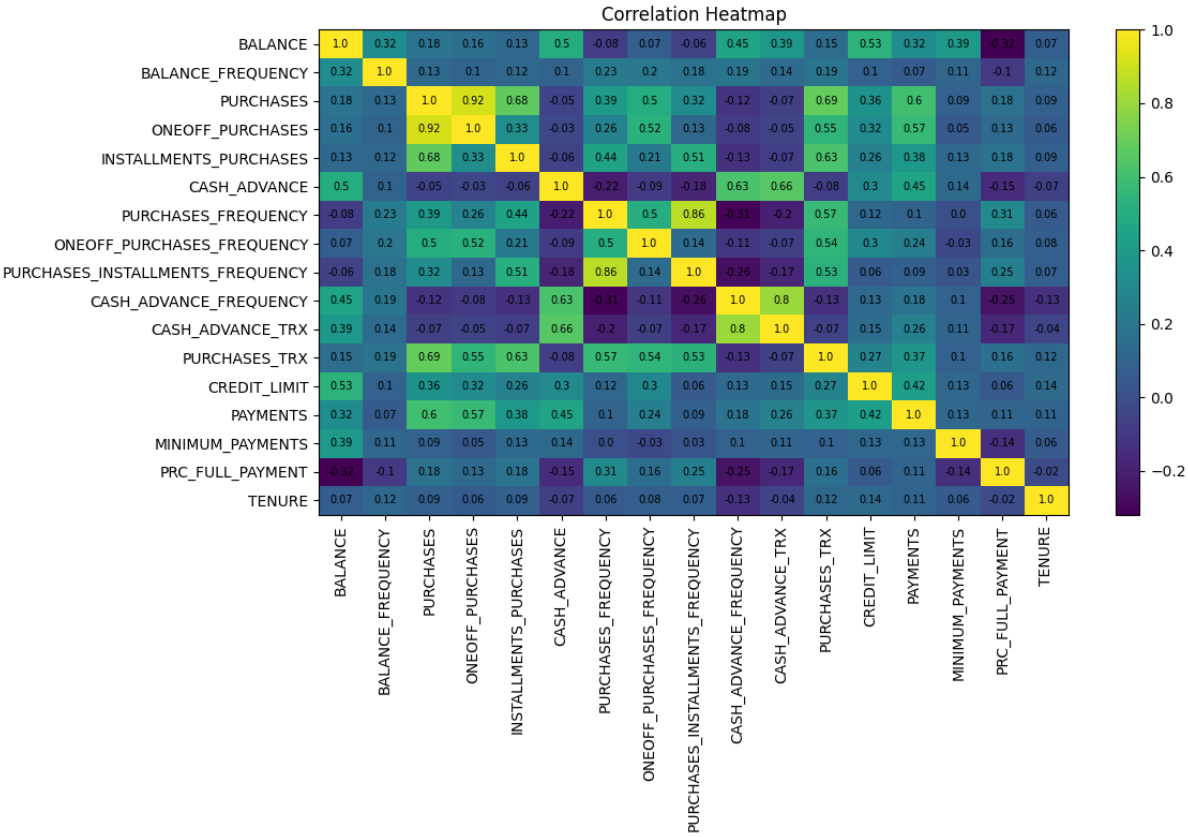
- Applied **Standard Scaler** to normalise the data
- Ensured all variables contribute equally to clustering

6.2 Correlation Analysis

A correlation heatmap was generated to understand relationships between features.

- Identified strongly related variables

- Helped in understanding behavioural dependencies

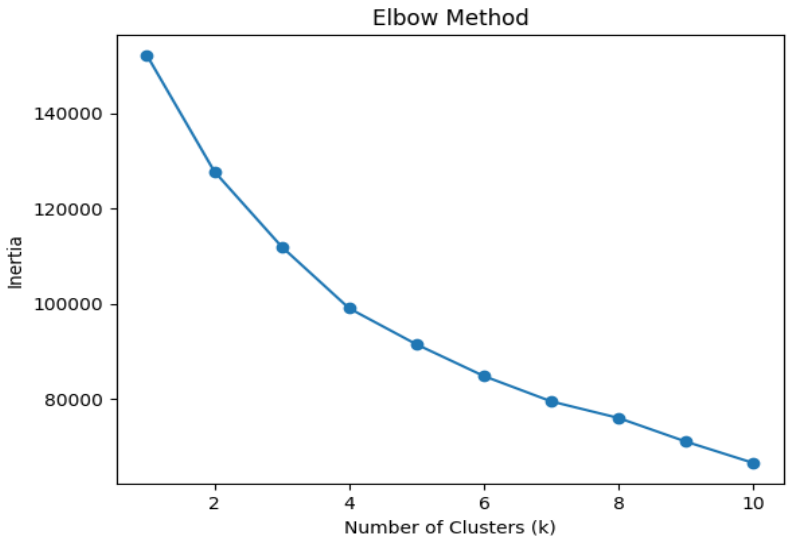


6.3 Determining Optimal Clusters

Elbow Method

- Plotted inertia vs number of clusters (k)
- Observed the "elbow point" to determine optimal k

  As the number of clusters increases, inertia decreases because data points are closer to



their cluster centroids. However, after a certain point, the rate of decrease slows down significantly, forming a bend or "elbow" in the graph.

This "elbow point" indicates the optimal number of clusters, as adding more clusters beyond this point provides only minimal improvement.

Silhouette Score

- The Silhouette Score is used to measure how well the clusters are separated and how closely related the data points are within the same cluster. It ranges from **-1 to 1**, where a higher value indicates better clustering.
- Higher score indicates better-defined clusters

Result:

Based on the silhouette scores, the optimal number of clusters was selected as **k = 3**, as it provided the best balance between cluster cohesion and separation.

6.4 K-Means Clustering

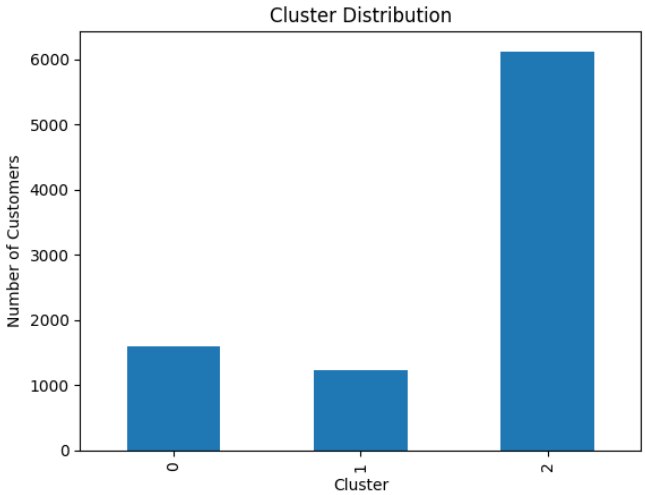
- Applied K-Means with **3 clusters**
- Assigned each customer to a cluster

6.5 Dimensionality Reduction (PCA)

- Used **Principal Component Analysis (PCA)** to reduce data to 2 dimensions
- Enabled visual representation of clusters

7. Results and Analysis

7.1 Cluster Distribution



- Each cluster represents a different behaviour group

- Customers are distributed across 3 clusters

7.2 Cluster Characteristics

Based on feature averages:

Cluster 0 – Low Activity Users

- Low purchases
- Low balance
- Low payments
- Likely minimal credit usage

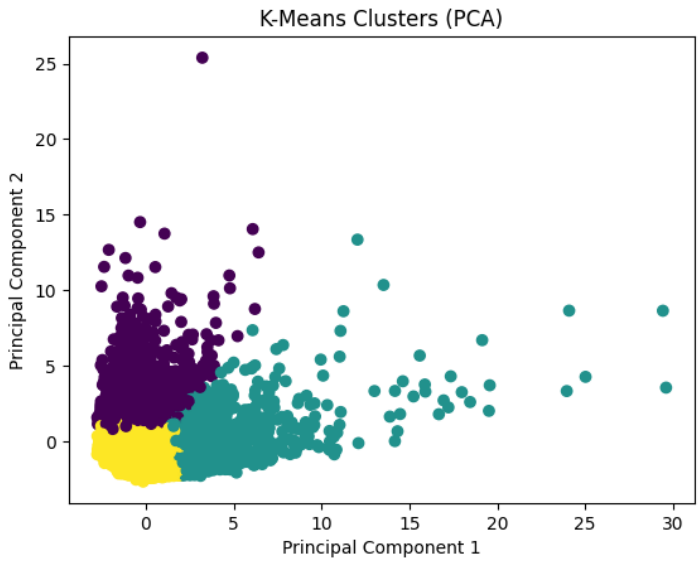
Cluster 1 – High Value Customers

- High purchases
- High credit limits
- High payments
- Premium or highly engaged users

Cluster 2 – Moderate Users

- (moderate) purchases and balance
- payments
- Regular but controlled usage

7.3 PCA Visualisation Insights



- Boxplots highlight differences in financial behaviour

- Distinct patterns confirm meaningful segmentation

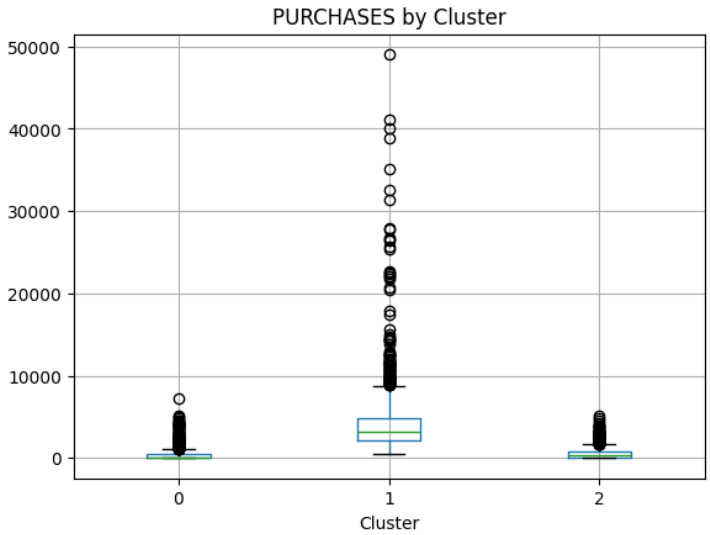
8. Model Evaluation

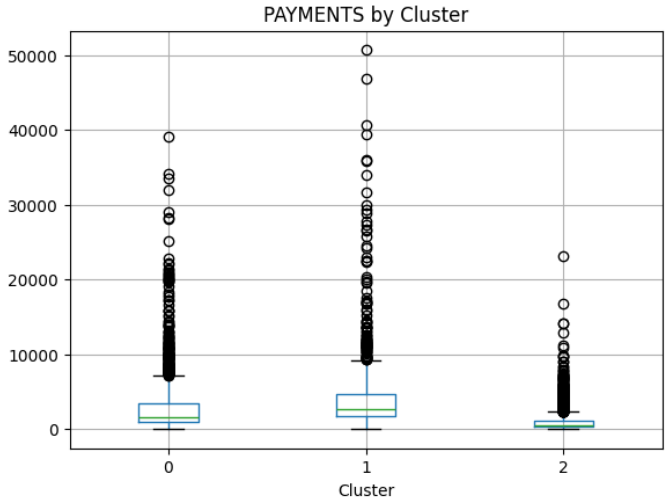
- **Silhouette Score** used to evaluate clustering performance
- Final score indicates reasonably well-separated clusters

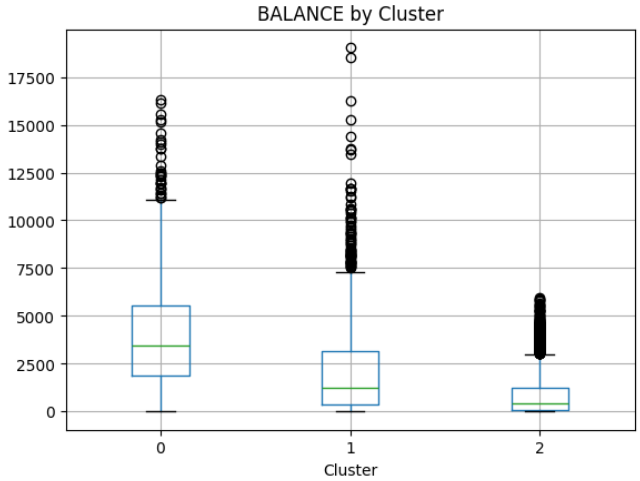
Interpretation:

- Score closer to 1 → strong clustering
- Score near 0 → overlapping clusters

Cluster Interpretation Plots







9. Business Implications

The identified clusters can be used for:

- **Targeted Marketing**

    - Offer premium services to high-value customers
    - Encourage low-activity users to increase engagement
- **Risk Management**
    - Monitor high-balance users
    - Detect unusual financial patterns
- **Customer Retention**
    - Personalised offers based on behaviour

10. Advantages and Limitations:

- **Dimensionality reduction (PCA):** Simplifies many variables (spending, payments, usage) into fewer components
- **Faster clustering: PCA reduces data size** → K-means runs quicker
- **Removes noise & correlation**: Improves cluster quality
- **Better visualization:** Easy to plot customer segments in 2D/3D
- **Efficient for large datasets:** Works well in real credit card systems

11.Future Work

- Experiment with different clustering algorithms (e.g., DBSCAN, Hierarchical Clustering)
- Perform feature engineering for better segmentation Use more advanced dimensionality reduction techniques
- Integrate clustering into real-time recommendation systems

12. Conclusion

This project demonstrates that K-Means clustering is an effective technique for customer segmentation.

Key takeaways:

- Customers can be grouped into meaningful behavioural segments
- Clustering helps uncover hidden patterns in financial data
- Results can drive strategic business decisions

13. References

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